Assignment — Unit 4 & Unit 5

Unit 4: Information Architecture (Labeling & Searching) | Unit 5: Enterprise Architecture (TOGAF 9)

Unit 4 — Labeling Systems & Search

12 questions

Q1 Explain the concept of labeling systems. Why are they essential in organizing and retrieving information on websites and intranets?

- **Label** = a short word/phrase/icon that **represents a larger chunk of information** (e.g., "Contact Us").
- **Labeling system** = a **consistent set of labels** used across navigation, headings, links and index terms of a site.
- **Why essential:** it lets users **predict what they will find** without opening pages (information scent).
- Reduces **cognitive load** and saves screen space — users scan instead of read.
- **Consistency builds trust** and makes navigation learnable; it also powers **search, browse and index terms**.
- On **intranets** every department uses its own jargon, so a common labeling system avoids confusion and lost productivity.
- **Poor labels = lost users**: people cannot find content even when it exists.



Q2 Differentiate between "labeling systems" and "labels." How do they work together to improve navigation and understanding?

Basis	Label	Labeling System
Meaning	A **single** term/icon for a piece of content	**Organised set** of labels with rules
Scope	One link, heading or menu item	Whole site / intranet
Example	"Products"	Home, Products, Support, Contact — same style & granularity everywhere
Focus	Meaning of one item	**Consistency** of syntax, style, granularity, audience

- **Together:** labels are the **words**, the system is the **grammar** that keeps them predictable.
- Result: users **learn once, apply everywhere** → faster navigation and better understanding of site structure.

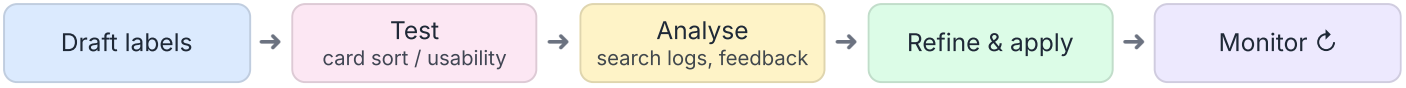
Q3 Describe the various types of labeling systems with examples, advantages and limitations.

Type	Example	Advantages	Limitations
Contextual links	Hyperlinks inside text (Wikipedia)	Used in context; flexible	Depends on author; inconsistent, ambiguous
Headings	"Products", H1/H2 titles	Shows hierarchy; easy scanning	Must match content; poor hierarchy confuses
Navigation choices	Menu: Home, About, Contact	Consistent, short, predictable	Limited space; vague words ("Resources")
Index terms	Keywords/tags: "Cloud computing"	Precise search & browse; controlled vocabulary	Costly to build/maintain; vocabulary drift
Iconic labels	Cart, Home, Search icons	Save space; language-independent	Ambiguous; limited vocabulary; need text support

Q4 What are the key principles for creating effective labeling systems? Discuss with examples from well-structured websites.

- **Narrow the scope** – design for a specific audience and content set.
- **Be consistent** in **style, syntax, presentation, granularity, comprehensiveness**.
- **Use the user's language**, not internal jargon (Amazon: "Cart", "Your Orders").
- **Follow conventions**: "Home", "Contact Us", "Search" (BBC, GOV.UK).
- **One term = one meaning**; avoid ambiguity and overlapping labels.
- **Be short and clear**; use verbs for actions ("Download", "Sign in").
- **Gather sources:** existing content, authors, search logs, card sorting, competitors.
- **Test with real users** before finalizing.

Q5 Explain the process of fine-tuning a labeling system. Why is continuous improvement necessary?



- **Steps:** build a draft list → create an **authority file / controlled vocabulary** → test with users → analyse results → revise → apply consistently → keep monitoring.
- **Why continuous:** content keeps **growing**, user language **changes**, new audiences and products appear.
- Mergers / reorganisations introduce **new terminology**; without upkeep labels drift and become inconsistent.
- Regular review keeps the site **usable, relevant and findable**. A labeling system is **never "finished"**.

Q6 What are non-representational labeling systems? Discuss their characteristics and provide examples.

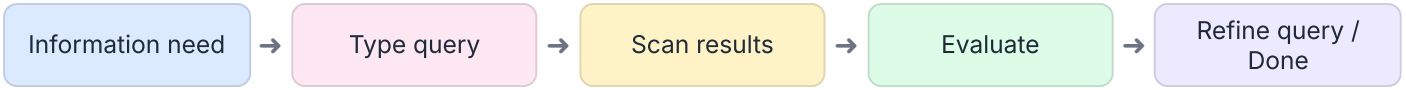
- **Definition:** labels that **do not describe the content in plain words**; they work through symbols, brand names or learned conventions.
- **Characteristics:** abstract/symbolic, **memorable**, brand-driven, **must be learned**, weak "information scent" at first sight.
- **Examples:** icons (hamburger ☰ menu, cart), brand/product names (**Gmail, Kindle, Slack**), codes/abbreviations (SKU, FAQ), creative names ("Sandbox", "Hub").
- **Advantage:** unique, compact, strong branding.
- **Limitation:** new users may not understand → best paired with a **text label or tooltip**.

Q7 Discuss the "double challenge" in designing labeling systems. How can designers overcome it?



- **Challenge 1:** language is **ambiguous** – one word, many meanings (polysemy) and many words, one meaning (synonymy).
- **Challenge 2: users and designers think differently**; different audiences, departments and contexts use different terms.
- **Solutions: card sorting** and user testing; **controlled vocabulary / thesaurus** with synonyms.
- Write a **style guide** (case, syntax, granularity); involve content owners and stakeholders.
- Use **search logs** to learn real user words; review and revise regularly.

Q8 Explain how users search websites. What behavioral patterns influence search system design?



- **Search modes:** **known-item** (I know what I want), **exploratory**, **exhaustive research**, and **re-finding**.
- **Short queries** (2–3 words); users rarely use **Boolean/advanced** search.
- Users rarely go beyond the **first page** of results; they **satisfice** (pick the first good-enough result).
- **Berrypicking:** needs evolve and users **reformulate queries** repeatedly; typos are common.
- They **search when lost** in navigation and expect **Google-like** simplicity.
- **Design impact:** simple box, **spell check, autocomplete, synonyms**, strong ranking, easy refinement.

Q9 Describe the major considerations in designing an effective search interface. How does it affect efficiency and satisfaction?

Consideration	Good practice	Effect
Search box	Visible at top, wide enough, labeled “Search”	Easy to find & use
Scope	Clearly show what is searched (site / section)	Fewer irrelevant hits
Assistance	Autocomplete, “Did you mean”, hints	Fewer errors, faster queries
Results display	Title, snippet, highlighted terms, metadata, count	Quick relevance judgement
Ranking & sorting	Most relevant first; sort by date/price	Right result near top
Refinement	Filters / facets, “search within results”	Narrows large result sets
No-result page	Suggest alternatives, tips, contact	Avoids dead end

- **Overall:** good interface design cuts **search time and effort**, boosts **confidence** and user satisfaction; poor design makes even a good engine useless.

Q10 What is a reference interview? How does understanding users' information needs improve search and labeling?

- **Reference interview:** a librarian's **structured conversation** with a patron to uncover the **real information need** behind the first question.
- It asks: **what, why, how much, in what format, by when?** – the stated question is often not the real need.
- **On the web:** done via **user interviews, surveys, personas, search-log analysis** and stakeholder discussions.
- **Benefits:** labels written in **users' own words**; search tuned with **synonyms and best bets**.
- Reveals **content gaps** and frequent tasks → better navigation, relevance and satisfaction.

Q11 Explain “indexing the right stuff.” Why is proper indexing critical?

- **Meaning:** deciding **which content and which parts** (title, abstract, full text, metadata) to make searchable – **not everything**.
- **Choose** high-value content; **exclude** outdated, duplicate and navigation-only pages.
- Create **search zones** (Products, Support, News) so users can limit scope.
- Index **metadata** (title, author, date, subject) with **controlled vocabulary**.
- **Why critical:** improves **precision** (relevant results) and **recall** (nothing missed), speeds up search, supports ranking.
- **“Garbage in, garbage out”:** poor indexing destroys user trust in search.

Q12 Discuss the dilemma of "to search or not to search." When should search be provided and when should navigation suffice?

Provide search when...	Navigation alone is enough when...
Site is **large** with many pages/silos	Site is **small** and well organised
Content is **dynamic** / frequently updated	Content is **stable** and homogeneous
Users look for **specific known items** (e-commerce, intranet docs)	Tasks are simple and **linear**
Resources exist to **tune and maintain** search	No resources to maintain – **bad search is worse than none**

- **Dilemma:** search is expected by users, but needs **setup, tuning and upkeep**; a poor search frustrates.
- Search and navigation are **complementary** – never use search as a **bandage for bad information architecture**.

Unit 5 — TOGAF 9

12 questions

Q1 Explain the concept and importance of TOGAF 9.

Business Architecture → Data Architecture → Application Architecture → Technology Architecture

TOGAF covers four architecture domains of the enterprise

- **TOGAF** = **The Open Group Architecture Framework**; an **open, vendor-neutral** framework for **Enterprise Architecture (EA)**. Version 9 released in 2009 (modular structure).
- Provides a **method (ADM)**, content framework, reference models and guidelines to **design, plan, implement and govern** enterprise architecture.
- **Importance:** aligns **business and IT**; gives a **common language**; reduces **cost and risk**.
- Promotes **reuse**, avoids **vendor lock-in**, improves **ROI** and supports **governance**.
- Widely accepted **industry standard** with certification.

Q2 Write a detailed note on the TOGAF 9 Management Overview.

Business strategy → Enterprise Architecture → Projects & solutions → Business value

- **Purpose:** explains to **executives and managers** why EA is needed and what it delivers.
- **Business drivers:** growing IT complexity, cost pressure, need for agility and **business–IT alignment**.
- **Benefits:** more efficient **business and IT operations**, better decisions, **lower risk**, better ROI, faster change.
- **Management role:** sponsor the effort, set up the **architecture capability and Architecture Board**, link EA to strategy.
- **Key points:** start with business goals, deliver in **iterations**, enforce **governance**, measure outcomes.

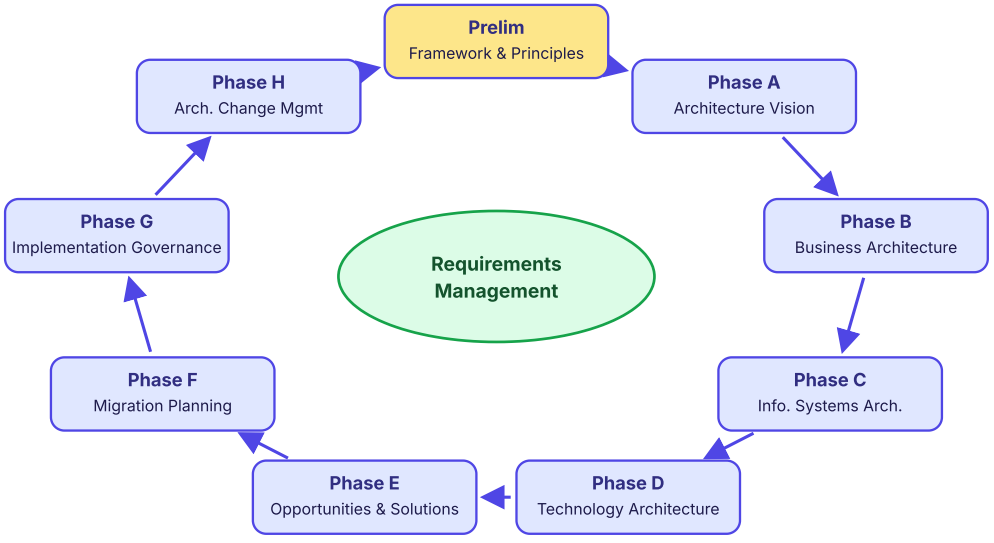
Q3 Describe the major components of TOGAF 9.

Part I Introduction	**Part II** Architecture Development Method (ADM)	**Part III** ADM Guidelines & Techniques	**Part IV** Architecture Content Framework
Part V Enterprise Continuum & Tools	**Part VI** TOGAF Reference Models	**Part VII** Architecture Capability Framework	

- **ADM** is the **core**; Content Framework defines **deliverables/artifacts**; Continuum & Repository support **reuse**.
- Reference models: **TRM** (Technical Reference Model) and **III-RM** (Integrated Information Infrastructure).
- Capability Framework = **organisation, processes, skills, roles** to run EA.

Q4 What is the Architecture Development Method (ADM)?

- **ADM** = a **step-by-step, iterative method** to develop and manage the **lifecycle of enterprise architecture**; the **core of TOGAF**.
- Consists of **Preliminary phase + Phases A–H** with **Requirements Management** at the centre.
- **Iterative** – across the whole cycle, between phases and within a phase.
- **Tailorable** to the organisation; **driven by requirements** and uses the Enterprise Continuum for reuse.
- Produces **deliverables** at each phase to ensure traceability.



TOGAF ADM cycle — iterative, with Requirements Management at the centre driving every phase

Q5 Explain all the phases of ADM.

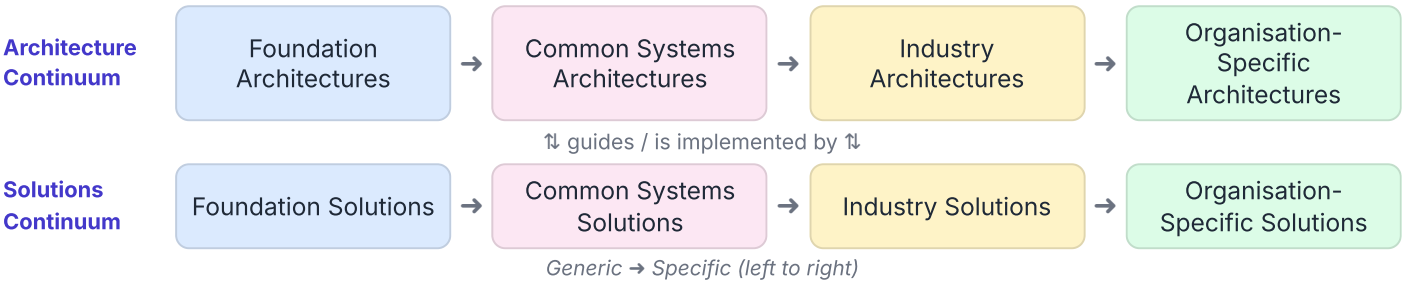
Phase	Purpose / Activities	Key Output
Preliminary	Prepare organisation; set **principles, governance**, tailor framework	Architecture principles, capability
A – Vision	Define **scope, stakeholders, vision, business case**	Statement of Architecture Work
B – Business	Baseline & target **business architecture**; gap analysis	Business architecture
C – Info. Systems	**Data** and **Application** architectures	Data & application architecture
D – Technology	Infrastructure, platforms, networks	Technology architecture
E – Opportunities & Solutions	Identify **work packages**, projects, transition architectures	Draft Architecture Roadmap
F – Migration Planning	Prioritise projects; cost-benefit & risk; **migration plan**	Implementation & Migration Plan
G – Implementation Governance	Oversee implementation; **compliance reviews**	Architecture Contract
H – Change Management	Monitor changes; handle change requests; start new cycle	Change requests, updated architecture
Requirements Mgmt	**Continuous** capture and control of requirements in all phases	Requirements Specification

Q6 Define ADM Deliverables and their significance.



- **Deliverables** = formal **work products** produced by ADM phases, reviewed and **agreed with stakeholders**.
- **Examples:** Architecture Vision, **Statement of Architecture Work**, **Architecture Definition Document**, **Architecture Requirements Specification**, **Architecture Roadmap**, **Architecture Contract**, Implementation & Migration Plan, Compliance Assessment.
- **Significance:** ensure **traceability** between phases; **formal sign-off** and stakeholder agreement.
- Act as **input to the next phase**, provide evidence for **governance**, enable **reuse** and measure progress.

Q7 What is the Enterprise Continuum in TOGAF 9?



- **Definition:** a **classification model** that organises architecture and solution assets from **generic to organisation-specific**.
- Has two parts: **Architecture Continuum** (what is required) and **Solutions Continuum** (how it is implemented).
- **Purpose:** promotes **reuse**, helps locate suitable assets, shows how architectures are **tailored** and **communicates** the relationship between assets.

Q8 Explain the Architecture Repository.

Architecture Metamodel structure & content rules	**Architecture Capability** parameters, structures, processes	**Architecture Landscape** Strategic / Segment / Capability views
Standards Library approved standards	**Reference Library** guidelines, templates, patterns	**Governance Log** decisions, compliance, dispensations

- **Definition:** a **structured store** holding all architecture outputs and related assets of the enterprise.
- Supports **consistency, reuse, traceability** and a single source of truth; it is the **physical storage** of items classified by the Enterprise Continuum.

Q9 Discuss the importance of Architecture Governance.

- **Definition:** the practice by which enterprise architectures are **managed and controlled** at enterprise level.
- Ensures **compliance** with standards, principles and the architecture vision.
- **Controls risk**, keeps projects consistent and prevents **architecture drift**.
- Brings **accountability**, clear decision rights and links to **corporate and IT governance**.
- Tools: **Architecture Board, Architecture Contracts, compliance reviews, dispensations**.
- Principles: **discipline, transparency, independence, accountability, responsibility, fairness**.
- Result: **business value is realised**, change is controlled, cost is reduced.

Q10 Define Views and Viewpoints.

Stakeholders	→	Concerns	→	**Viewpoint** template / perspective	→	**View** actual representation

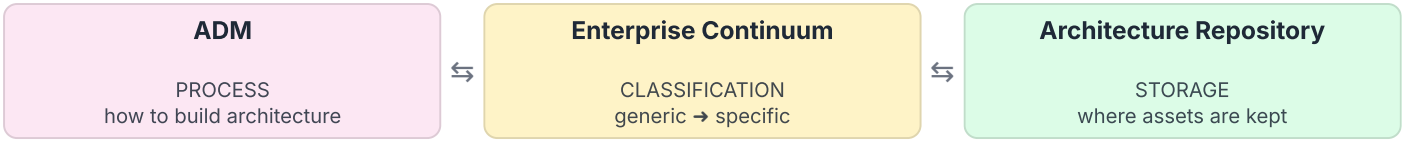
- **View:** a representation of a system from the perspective of a **related set of concerns** – **what** is seen.
- **Viewpoint:** a specification of the **conventions and perspective** used to build a view – **how** it is seen.
- Based on **ISO/IEC 42010** (IEEE 1471); each **stakeholder** has different concerns (security, cost, performance).
- **Example:** *Security viewpoint → security architecture diagram* for the security engineer. Like a building: electrical plan for the electrician, floor plan for the owner.
- Ensures **all stakeholder concerns** are addressed and communicated properly.

Q11 What are Architecture Building Blocks (ABBs)?

ABB what is needed (architecture level)	→	**SBB** how it is implemented (product / solution)

- **ABB:** a **reusable, technology-neutral** component of business, IT or architectural capability that defines **required functionality**.
- Captures **requirements, interfaces and standards**; guides the selection of **Solution Building Blocks (SBBs)**.
- **Example:** ABB = "Authentication service" → SBB = a specific vendor identity product.
- Defined in **Phases A–D** (conceptual → detailed); SBBs are selected in **Phases E–F**.
- **Benefits:** **reuse**, flexibility, easier integration and vendor independence.

Q12 How do ADM, Enterprise Continuum, and Architecture Repository integrate?



- **ADM** pulls reusable **reference assets** from the Repository as inputs (e.g., standards, patterns).
- The **Enterprise Continuum** **classifies** these assets so the right level (foundation → organisation-specific) is chosen.
- ADM **outputs** (deliverables, artifacts, building blocks) are **stored back** in the Repository and classified in the Continuum.
- Specific assets are generalised for reuse; generic ones are **tailored** – creating a **continuous reuse loop** under **governance**.
- **In short:** ADM = *process*, Continuum = *context*, Repository = *storage* → together they give consistent, repeatable architecture.